

AJAY BHATTARAI

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Biratnagar, Koshi Province-56613, Nepal

PROFESSIONAL SUMMARY

Civil Engineer with a strong academic background and research interests in **Hydrology, Water Resources, Climate Change Impacts, Groundwater, floods, Drought**. Experienced in **QGIS**, numerical modeling with **Python**, contributing to national-level water resources projects. Highly motivated to apply advanced **Machine Learning (ML) and Deep Learning (DL)** and develop the tools to solve complex hydrological and climate-driven water challenges.

EDUCATION

- **Institute of Engineering (IOE), Pulchowk Campus, Tribhuvan University** 2024
Bachelor of Civil Engineering Lalitpur, Nepal
 - PERCENTAGE: 77.55/100%
 - IOE Rank: 174 (Top 1% of national applicants)
- **Kantipur College** 2019
Higher Secondary Education Biratnagar, Nepal
 - GPA: 3.59/4 (**Batch Topper**)
- **Pokhariya Secondary School, Biratnagar** 2017
Secondary Education Examination (SEE) Biratnagar, Nepal
 - GPA: 3.75/4 (**Batch Topper**)

PROFESSIONAL RESEARCH EXPERIENCE

- **Research Assistant | Center for Applied Research and Development (CARD)** January 16, 2026 - Present
Tribhuvan University, Lalitpur, Nepal
 - **Bagmati River Basin Plan (Asian Development Bank & WECS):**
 - Assisting in the design of future basin-wide water development scenarios to establish a national water security strategy updated for the year 2050.
 - Processing climate, population, hydropower, and irrigation data and projecting future water demand under different scenarios.
 - Contributing to water balance and demand analysis for irrigated agriculture and drinking water to evaluate future climate change impacts.
 - **Chure Region River System Atlas (PCTMCDB):**
 - Delineated and classified 220 hydrologically consistent river systems (32 watersheds and 188 sub-watersheds) to support the national Chure Conservation Master Plan.
 - Analyzed and corrected river network datasets using ALOS AW3D30 digital elevation models (DEM), stream burning techniques, and QGIS terrain analysis.
 - Contributed to the technical writing of the final atlas report and prepared publication-ready maps.
 - Developed a React-based **web application** to digitally visualize the project's river systems, maps, and spatial data. ([App Link](#))
- **Reserach on Groundwater Dynamics Project** 2025
Main Author | Currently Under Review
 - Assessed the current status of groundwater and identified key drivers of depletion
 - Examined the effects of rainfall, shallow and deep tube wells, land use/land cover (LULC) changes, and policies on groundwater dynamics
 - Analyzed climate data (rainfall, temperature) and their influence on groundwater recharge
 - Performed spatio-temporal analysis using the insitu data
 - Developed research-based recommendations for sustainable groundwater resource management frameworks.

TECHNICAL & AI PROJECTS

- **Machine Learning & Deep Learning Applications in Engineering** 2025
Python, PyTorch, YOLOv8, OpenCV, ANN, CNN
 - **Concrete Strength Prediction (ANN):** Developed a fully connected Artificial Neural Network regression model to predict compressive strength by analyzing 7 key mix-design parameters. [\(App Link\)](#)
 - **Bidirectional Vehicle Counting (YOLOv8):** Built a real-time Deep Learning system using YOLOv8 and ByteTrack for high-accuracy detection and tracking of multi-class vehicles. [\(GitHub\)](#)
 - **Structural Crack Detection (CNN):** Designed and trained a custom Convolutional Neural Network using PyTorch for automated, binary classification of infrastructural surfaces. [\(GitHub\)](#)
- **Civil Engineering Infrastructure Projects** 2024
Civil 3D, SW-ROAD, AutoCAD, MS Excel
 - **Road Design & Hydrology:** Conducted detailed engineering surveys, alignment design, and 3D visualization. Calculated maximum drainage discharge and designed drainage and retaining wall facilities (IS 14458-1).
 - **Grading & Estimation:** Performed grading, alignment design, and material take-offs for hydropower and infrastructure projects in Dolakha.

TEACHING & TRAINING EXPERIENCE

- **Engineering Software & Mechanics Instructor** Oct 2023 - Present
Pulchowk Campus, GyanHub Academy, Sikaune EduTech, Universal Engineering College
 - Trained 500+ students and professionals in AutoCAD, Revit, SW-DTM, and Google Earth Pro from basic to advanced levels across multiple institutions.
 - Taught comprehensive modules on contour creation, spatial data collection, and road alignment design.
 - Developed and currently delivering an online/recorded curriculum for Engineering Applied Mechanics and Engineering Drawings.

LEADERSHIP & EXTRACURRICULARS

- **President, Society of Eastern Engineering Students (SOEES):** Led a major student organization and coordinated regional technical workshops (July 2023 - July 2024).
- **Program Coordinator:** Managed an inter-college AutoCAD competition (10+ colleges) for CESS and oversaw logistics for the 32nd Grand Saraswati Puja for NTBNS.
- **Team Leader:** Led the final year civil engineering project team for a detailed topographic survey and road design.
- **Secretary, Leo Club of Kathmandu Pashupatinath:** Managed club administration and community event planning.

SKILLS

- **Engineering & GIS:** AutoCAD, QGIS, Arc-GIS, Hec-Ras, GEE, Civil 3D, SW-DTM, SW Road
- **Programming & AI:** Python, Pandas, Numpy, Matplotlib, PyTorch, Streamlit, HTML, CSS, React
- **Documentation:** LaTeX (Overleaf), MS Office

ACHIEVEMENTS & ADDITIONAL INFORMATION

- **Awards:** Winner of 1st Inter-College AutoCAD Competition | Batch Topper (SEE & Higher Secondary)
- **Languages:** Nepali (Native), English (Fluent), Hindi (Conversational)
- **English Proficiency Test (Duolingo):** Overall: 120 (Each band not less than 110)

REFERENCES

Dr. Vishnu Prasad Pandey Professor <i>Direct Research Supervisor</i> IOE, Pulchowk Campus +977-9841318939 vishnu.pandey@pcampus.edu.np	Dr. Basanta Raj Adhikari Assistant Professor <i>Undergraduate Instructor</i> IOE, Pulchowk Campus +977-9851133510 bradhikari@ioe.edu.np	Dr. Suraj Lamichhane Professor <i>Co-PI, Chure Project</i> IOE, Pulchowk Campus +977-9851212144 klsuraj@gmail.com
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